

# NALINA GENKESWARAN

Undergraduate, Computer Science and Engineering,  
University of Moratuwa

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## ABOUT ME

Third-year Computer Science and Engineering undergraduate at the University of Moratuwa, passionate about backend development and building scalable solutions for real-world problems. A continuous learner who enjoys exploring new technologies, demonstrating leadership, collaborating with teams, and continuously improving technical and professional skills.

## EDUCATION

Bsc(Hons) in Computer Science and Engineering



Feb 2024 - Ongoing

**University of Moratuwa.**

CGPA - 3.45

GCE Advanced Level - 2022



2020 - 2022

**Vembadi Girl's High School, Jaffna.**

Z.Score - 2.2574

## PROJECTS

**An AI-Powered Technical Debt Analytics Dashboard for Agile Small-Scale Software Teams- In Progress**

**Spring Boot, React.js, Python, Tree-sitter, Lizard, PostgreSQL, RabbitMQ, Machine Learning**

- Developing an AI-powered SaaS platform for technical debt analytics and software repository analysis.
- Working as a Full-Stack Software Engineer, contributing to the frontend, backend, and AI-driven analysis modules.
- Building an intelligent repository analysis engine that detects technical debt, predicts defect-prone files, and generates prioritized refactoring recommendations.

**AgriSenseNet (CropWise) – Smart Agriculture Platform – Completed**

**Docker, Kubernetes, AWS, Terraform, GitHub Actions, ArgoCD, Keycloak, Prometheus, Grafana**

- Developed an IoT-enabled smart agriculture platform for crop monitoring and irrigation management.
- Served as a Platform, Security & Integration Engineer (C4 Team).
- Engineered a secure and scalable deployment platform by implementing infrastructure automation, deployment pipelines, authentication, and observability.

**Offline-First Emergency Communication & Disaster Prioritization System – In Progress**

**Spring Boot, PostgreSQL, React.js, Google Gemini API, SMSMobile API, REST APIs**

- Developed an offline-first emergency communication and disaster prioritization system for reliable disaster response.
- Contributed as a Backend Developer in a 4-member team.
- Developed backend services for AI-assisted disaster severity analysis, priority-based response, SMS integration, and volunteer matching.

Rescue Team Management System (Individual Project) – Completed

Spring Boot, Java, React.js, MySQL, Spring Data JPA, REST APIs

- Developed a full-stack rescue team management system to streamline disaster response and incident management.
- Built the application independently as a Full-Stack Developer.
- Implemented role-based workflows, REST APIs, and dashboards for emergency reporting, rescue team assignment, and incident tracking.

Microbanking & Interest Management System (MIMS) - Completed

Next.js, PostgreSQL, JWT, REST APIs, Tailwind CSS

- Developed a full-stack microbanking system for customer management, banking transactions, and financial operations.
- Contributed as a Full-Stack Developer in a 5-member team.
- Implemented secure authentication, RESTful APIs, responsive user interfaces, and automated interest distribution.

SKILLS

Technical

- **Programming Languages:** Java, Python, C++ , C, JavaScript/TypeScript, SQL
- **Frameworks & Libraries:** Spring Boot, React, Node.js, Express
- **Databases:** MySQL, PostgreSQL, , MongoDB
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Terraform
- **Tools & Technologies:** Git, GitHub, Kafka, Postman, Linux

VOLUNTEERING & LEADERSHIP

Frontend Developer | Mora Exams Community  2023, 2025

- Developed frontend features for the Mora Exams Website and Admin Dashboard supporting island-wide A/L examinations.
- Played an active role in the planning, coordination, and successful execution of large-scale examinations, working with cross-functional teams to manage operations and participant engagement across Sri Lanka.

Company Coordinator | CSE Career Fair, University of Moratuwa

- Coordinated with participating companies and supported communication between company representatives and the organizing team.

ACHIEVEMENTS

- **IAEA Nuclear Science App Challenge 2026** – Sri Lanka Representative (University Category Winner)
- **CryptX Hackathon 2026** – Finalist
- **UOJ Coders Hackathon 2026** – Finalist

CERTIFICATIONS

- Java Programming by simplilearn (December 2025)
- Intro to Machine learning by kaggle (January 2026)
- Data cleaning by kaggle (February 2026)
- Pandas by kaggle (February 2026)
- SQL by HackerRank ((December 2025)

REFEREES

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